

KELT-20b

R-1932

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-20_260702 · created 2026-08-06T13:37:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461223.8777 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.013 +/- 0.041
Transit depth (Rp/Rs)^2	0.016 +/- 0.1 [%]
Orbital Inclination (inc)	85.29 +/- 2.54
Airmass coefficient 1 (a1)	1.0273143 +/- 8.5e-06
Airmass coefficient 2 (a2)	-0.026 +/- 0.03
Scatter in the residuals of the lightcurve fit is	2.4e-05 %
Best Comparison Star	#1 - [302, 397]
Optimal Aperture	2.85
Optimal Annulus	16.4
Transit Duration (day)	0.041 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.013 ± 0.041 vs 0.1144 (deviation 2.47σ)
Duration fit vs published	0.041 d vs 0.1471 d (deviation 7.07σ)
Mid-time O-C	-5.9 min
Depth SNR	0.3
Residual scatter	2.4 %

Light curve

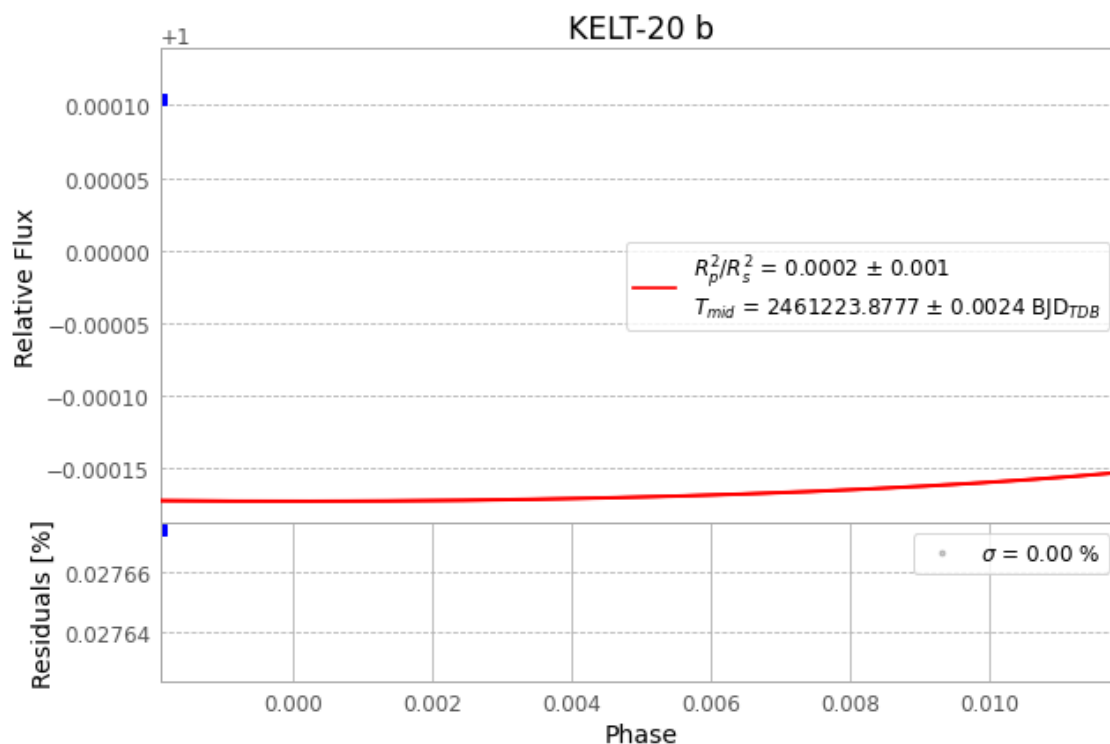


Figure 1 — FinalLightCurve_KELT-20 b_2026-07-02.png (SHA-256 34a1eb2dc43161be...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [487, 396], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c7ebcaa9ac1f0777...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 341f7e9

Data provenance

177 FITS frames (70 MB), KELT-20260702061628.FITS.gz → KELT-20260702122345.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-20-260702.log (18bc2dc25e13...), FinalLightCurve_KELT-20 b_2026-07-02.png (34a1eb2dc431...), FinalLightCurve_KELT-20 b_2026-07-02.csv (bd7eea82539b...)

Compute resources

Wall time 146.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 13:37Z.